Artemisia terpene diversity across scales: biosynthetic evolution and bounded antiparasitic evidence

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Abstract

The genus *Artemisia* contains volatile monoterpenes and sesquiterpenes, less volatile diterpenes and triterpenes, and structurally diverse sesquiterpene lactones (STLs). These metabolites are often discussed as though a species name, a pathway gene, or an oil profile were sufficient to predict pharmacology. Here we present a source-auditable living scoping review that instead treats the specimen, accession, tissue, preparation, analytical method, parasite stage, and evidence level as the units of comparison. The Terpedia package registers 538 sources, 553 source-linked evidence records, 129 antiparasitic records, 19 parasite–protein interaction records, 179 compound/specimen records, and 400 audited article claims; 79 sources and 94 linked evidence rows have completed full-text verification, while 31 queue sources remain open. We synthesize terpene classes with biosynthetic compartmentalization, regulatory and environmental variation, and a bounded comparative-genomics analysis. Artemisinin biosynthesis in *A. annua* is the best-resolved gene-to-metabolite benchmark, but it should not be used as a proxy for the genus. Malaria provides the strongest validated antiparasitic case: artemisinin has direct parasite-relevant mechanistic support, whereas non-artemisinin matrix and metabolomics studies mainly identify candidate contributors to preparation-level activity. Wormwood (*A. absinthium*) provides a separate vermifuge hypothesis supported by in-vitro nematode and limited animal evidence, but constituent attribution, host selectivity, and modern clinical translation remain unresolved. We propose a phylogeny-aware program linking voucher-authenticated metabolomics, secretory-tissue multi-omics, common reannotation, enzyme assays, and fractionated parasite testing. The result is a concise synthesis for journal submission with the complete evidence and provenance archive retained in Terpedia.

Keywords: *Artemisia*; artemisinin; wormwood; sesquiterpene lactones; malaria; comparative genomics

1. Introduction

Artemisia is a chemically and ecologically heterogeneous genus. Volatile profiles can change with genotype, population, tissue, developmental stage, harvest, drying, storage, extraction, and analytical platform. Nonvolatile terpenoids are subject to an additional set of constraints: solvent selectivity, tissue localization, thermal instability, stereochemistry, ionization behavior, and structural dereplication. The appropriate biological unit is therefore not "the *Artemisia* terpene profile," but a documented specimen or accession measured under a defined preparation and analytical context.

This distinction matters for antiparasitic interpretation. An essential oil is a mixture, an extract is a matrix, a purified compound is a defined exposure, and a transcript or homolog is a molecular hypothesis. These objects cannot be merged without losing the information needed to infer causality. The same problem appears in evolutionary genomics: a terpene-synthase (TPS) copy, an annotated cytochrome P450, or a shared orthogroup does not establish native product formation, pathway flux, or parasite activity.

Several important reviews already establish the breadth of the field. Essential-oil chemistry and anti-infective activity were synthesized at genus level by Abad et al. and updated for 2012–2017 by Pandey and Singh. Ivanescu et al. focused on STL analysis and biological activities; Trendafilova et al. reviewed edible *Artemisia* species and selected STLs; and Hussain et al. provided a broad review of traditional use, phytochemistry, pharmacology, and toxicology. Recent reviews of STL biosynthesis and evolution, *A. annua* production, antimalarial chemistry, and botanical anthelmintics further demonstrate that the literature is substantial rather than absent (Abad et al. 2012; Pandey and Singh 2017; Ivanescu, Miron, and Corciova 2015; Trendafilova et al. 2021; M. Hussain et al. 2024).

The gap addressed here is not another catalogue of compounds. It is an evidence model connecting four scales: (i) compound identity and abundance; (ii) biosynthetic enzymes, regulation, and compartment; (iii) phylogeny, genome organization, and chemotype variation; and (iv) parasite phenotype, target evidence, host controls, and translation. Malaria and artemisinin are used as the high-confidence anchor. Wormwood is treated as a distinct vermifuge case because historical santonin chemistry, *A. absinthium* preparations, and modern veterinary nematode assays are related but not interchangeable.

2. Review scope and evidence framework

2.1 Living scoping-review design

The review was organized around five domains: compound occurrence and analytical identity; pathway, enzyme, and compartment evidence; genotype, population, tissue, developmental, and environmental variation; comparative genomics and gene-family evolution; and antiparasitic phenotype, target, and safety evidence. Searches combined *Artemisia* terms with terpene classes, analytical methods, biosynthesis, genomics, malaria, helminths, vectors, and safety terms. The frozen retrieval snapshot contains 4,765 unique PubMed records from seven query families, of which 1,387 were prioritized for title-level review and 1,323 had abstracts retrieved. A source-linked full-text eligibility queue covers 110 unique antiparasitic sources representing 129 evidence rows, prioritizing malaria and *A. absinthium* vermifuge records.

The package is deliberately a living scoping review rather than a completed systematic review. It contains 1,408 bounded title/abstract decisions, covering all 1,387 priority-queue records plus 21 historical or out-of-queue decisions. Full-text verification now covers 79 publicly accessible or publisher-full-text antiparasitic sources and 94 linked evidence rows; 31 queue sources remain unverified, and no result is pooled as a final quantitative estimate. The PRISMA-ScR mapping is archived in `prisma-scr-checklist.md`; the search strings and retrieval counts are in `search-log.md` and `review-flow.md`. Reviews are used for discovery and context, while quantitative and mechanistic statements are preferentially anchored to primary studies.

Evidence was assigned according to what the source actually established. Structure-confirmed isolation or an authentic analytical standard supports compound identity in the tested material. A library-only GC-MS assignment is lower confidence. Heterologous enzyme activity supports catalytic capacity under assay conditions, not native flux. Transcript abundance, co-expression, and homology prioritize candidates but do not prove product identity. Parasite growth inhibition establishes a preparation-specific phenotype; causal target claims require biochemical, structural, genetic, or activity-based target engagement.

2.2 Chemical classes and evidence imbalance

Table 1 summarizes the chemical classes covered and the minimum fields needed for defensible comparison.

Table 1. Chemical classes and minimum interpretation requirements.

Class	Representative *Artemisia* chemistry	Typical evidence	Minimum comparison fields
Monoterpenes (C10)	α-/β-pinene, limonene, camphor, 1,8-cineole, borneol, thujones, artemisia ketone	GC-MS/GC-FID of oils or headspace; occasional isolation	Voucher, tissue, chemotype, stereochemistry, drying, retention index, standard
Sesquiterpenes (C15)	germacrene D, β-caryophyllene, α-humulene, davanone, spathulenol, caryophyllene oxide	Volatile profiling, isolation, enzyme assays	Preparation, abundance units, analytical confidence, tissue, stage
Diterpenes (C20)	species-specific diterpenoids and phytol-associated products	Solvent extraction, LC-MS, NMR, diTPS assays	Extraction polarity, structural confirmation, co-occurring matrix, native abundance
Triterpenes (C30)	amyrins, friedelin-, cycloartane- and sterol-associated chemistry	Solvent extraction, LC-MS, derivatized GC-MS, NMR, OSC assays	Surface/storage compartment, derivatization, standard, tissue and developmental context
Sesquiterpene lactones	artemisinin, arteannuin B, artemisinic acids, santonin, guaianolides, germacranolides	LC-MS/HPLC, NMR, isolation, enzyme and pathway studies	Stereochemistry, lactone structure, analytical standard, tissue, preparation, dose

Volatile mono- and sesquiterpenes dominate the searchable literature because routine GC methods make them accessible. Diterpenes, triterpenes, and STLs are less directly comparable because they require extraction, LC-MS, NMR, or targeted structural workflows. This is a sampling effect, not evidence that C10 and C15 chemistry is biologically more important. Conversely, a long list of isolated C20 or C30 compounds does not imply high abundance or antiparasitic relevance.

The under-sampled C20 and C30 classes nevertheless have a real, experimentally tractable literature. *A. annua* seed isolation recovered fourteen sesquiterpenes, three monoterpenes, and one diterpene, while a separate study identified ten *A. annua* diterpene-synthase genes in glandular-trichome and stress-resilience contexts (Brown, Liang, and Sy 2003; Chen et al. 2021). OSC2 and CYP716A14v2 were functionally linked to α-, β-, and δ-amyrin-related triterpenoids in the *A. annua* aerial cuticle (Moses et al.

2015). Structure-confirmed isolations extend this space to ent-kauranoids in *A. sacrorum*, a triterpene in *A. caruifolia*, an oleanane-type saponin in *A. sphaerocephala*, nordammarane and cycloartane triterpenoids in *A. argyi*, and new sesqui-/triterpenoids in *A. vulgaris* (X. Li et al. 1990; Ma et al. 2001; L. Li et al. 2008; L. Zhang et al. 2020; T. Liu et al. 2022). These records establish chemical space, not species-wide abundance or antiparasitic efficacy.

The less-volatile evidence anchors are summarized in Table 2.

Table 2. Evidence anchors for less-volatile terpene classes.

Layer	Representative evidence	What it supports	Boundary
C20 occurrence	*A. annua* seeds; *A. sacrorum* ent-kauranoids	Diterpene chemical space and isolation-level identity	Tissue- and extraction-bounded; no antiparasitic inference
C20 pathway	Ten *A. annua* diTPS genes	Candidate synthase diversity and stress-linked regulation	Copy number and expression do not prove product flux
C30 compartment	*A. annua* OSC2/CYP716A14v2 in aerial cuticle	Tissue-resolved triterpene biosynthesis	Native product and parasite function remain separate questions
C30 occurrence	*A. caruifolia*, *A. sphaerocephala*, *A. argyi*, *A. vulgaris* isolations	Cross-species scaffold diversity	Isolated compounds are not chemotype means
C30 catalytic evolution	*A. argyi* AaOSC-22030 mutagenesis	Active-site determinants of product plasticity	Recombinant/engineered specificity is not native selection

The *A. argyi* AaOSC-22030 study provides the strongest sequence-to-function example for C30 evolution: mutagenesis at S728 altered product specificity, and S728Y yielded dammarendiol II as the reported exclusive product, while N369/E371 influenced terminal hydration (C. Liu et al. 2026). This is a useful comparative experiment, but it does not establish native abundance, lineage-wide selection, or parasite activity.

3. From precursor supply to chemical phenotype

3.1 Pathway architecture and compartment

The plastidial MEP and cytosolic mevalonate pathways supply IPP and DMAPP, which are condensed into prenyl diphosphates. GPPS-related activity supplies C10 substrates for monoterpene synthases, while FPPS supplies FPP for C15 sesquiterpene synthases. C20 and C30 chemistry depends on additional prenyltransferase and cyclase branches. TPS enzymes create hydrocarbon skeletons; P450s, reductases, dehydrogenases, oxidases, acyltransferases, glycosyltransferases, and spontaneous reactions generate the oxygenated and rearranged products detected in plants.

The pathway is spatially organized. Glandular secretory trichomes can combine precursor supply, TPS activity, oxidation, storage, and chemical defense in a restricted tissue. In *A. annua*, immunolocalization and trichome transcriptomics placed key artemisinin-pathway activities in secretory trichomes, while tissue-expression studies showed that terpene-associated transcripts vary among roots, stems, leaves, and reproductive tissues (Olsson et al. 2009; W. Wang et al. 2009; Olofsson et al. 2011). A metabolite detected in bulk leaf is therefore not necessarily produced uniformly across the leaf, and bulk expression can obscure the cells that control accumulation.

Secretory capacity is itself regulated. AaMIXTA1 perturbation altered glandular-trichome initiation, cuticle-related transcription, and artemisinin content; AaPDR3 affected β-caryophyllene accumulation in T-shaped trichomes; and promoter-reporter work showed age- and tissue-dependent *CYP71AV1* expression (P. Shi et al. 2018; Fu et al. 2017; H. Wang et al. 2013). These results support a model in which chemical evolution can act through cell type, transport, storage, and cuticle architecture in addition to enzyme sequence.

3.2 Artemisinin as a benchmark, not a genus-wide proxy

Artemisinin biosynthesis in *A. annua* provides the strongest gene-to-metabolite reference in the genus. ADS converts FPP to amorpha-4,11-diene; CYP71AV1 oxidizes pathway intermediates; DBR2 and ALDH1 support the route toward dihydroartemisinic acid; and terminal chemistry includes nonenzymatic steps. Early cloning, recombinant assays, trichome-enriched expression, and isotope-feeding experiments established multiple pathway steps independently (Bouwmeester et al. 1999; Chang et al. 2000; Teoh et al. 2006; Y. Zhang et al. 2008; Schramek et al. 2010).

The regulatory evidence is similarly layered. AaWRKY1, AaNAC1, AaORA, AaMYB108, jasmonate-linked ERF factors, cold-responsive AabHLH112, miR160–ARF1, and post-translational AaPP2C1–AaAPK1 regulation have each been connected to pathway expression or metabolite accumulation through perturbation, promoter, interaction, or expression experiments (Jiang et al. 2016; Lv et al. 2016; Lu et al. 2013; Xiang et al. 2019; Z. Guo et al. 2023; T. Zhang et al. 2019). These studies establish causal regulation within tested *A. annua* backgrounds. They do not demonstrate conserved function in *A. absinthium*, *A. argyi*, *A. herba-alba*, or any other congener.

The same caution applies to production engineering. Recurrent selection increased artemisinin in *A. annua*, CYP71AV1/CPR co-expression increased production in transgenic lines, and a 2025 AaDHAADH variant improved dihydroartemisinic-acid production in engineered yeast (Paul et al. 2010; Shen et al. 2012; Zizheng Guo et al. 2025). These are strong interventions for production, but increased plant titer is not equivalent to increased parasite selectivity, bioavailability, or clinical efficacy.

3.3 Evidence for other terpene classes

Monoterpene synthase assays show that closely related enzymes can generate distinct product spectra. In *A. annua*, AaTPS2, AaTPS5, and AaTPS6 differed in recombinant products and responded to wounding and hormone treatments (Ruan et al. 2016). *A. annua* seed isolation recovered sesquiterpenes, monoterpenes, and a diterpene, while trichome work functionally characterized OSC2 and CYP716A14v2 in aerial-cuticle triterpenoid production (Brown, Liang, and Sy 2003; Moses et al. 2015). These records anchor C10, C15, C20, and C30 chemical space but do not define a genus-wide repertoire.

STLs are particularly important because the α-methylene-γ-lactone or related electrophilic functionality can react with nucleophilic protein residues. Yet structural class does not predict a single biological outcome. Guaianolides, germacranolides, eudesmanolides, artemisinin-type endoperoxides, and santonin-like compounds differ in stereochemistry, oxidation, ring fusion, reactivity, stability, and exposure. A structure-confirmed STL should therefore enter the evidence map first as a chemical occurrence; antiparasitic function requires a defined assay and, ideally, a matched host counter-screen.

4. Chemotype, tissue, environment, and analytical context

Across *Artemisia*, chemotype is an emergent phenotype rather than a fixed species label. Population, geography, phenology, plant age, light, temperature, drought, nutrient status, wounding, hormone treatment, and microbial association can alter both total production and the relative balance among branches. In *A. frigida* and *A. absinthium*, population and geography alter volatile composition; in *A. annua*, LED spectrum, plant density, soil copper, hydrogen sulfide, nitric oxide, and developmental stage alter artemisinin or volatile outputs (Zhigzhitzhapova et al. 2023; Judzentiene and Garjonyte 2016; Ram et al. 1997; Zehra et al. 2020; Nomani et al. 2022; K. I. Wani et al. 2023). These effects are biologically informative but are not automatically transferable as cultivation recommendations or parasite interventions.

Postharvest chemistry is a major confounder in antiparasitic comparisons. Extended shade drying changed *A. monosperma* oil yield and composition, while hot-air drying/fixation in *A. argyi* reduced selected oxygenated monoterpenes and increased selected sesquiterpenes (Mohammed and Alfazi 2026; Ming and collaborators 2025). Wormwood profiles also change through evaporation and parent-to-daughter transformations during drying and storage (Blagojević, Radulović, and Skropeta 2015). An assay using "dried wormwood" without temperature, duration, storage, oil yield, and chromatographic composition is therefore difficult to compare with an assay using fresh plant, tincture, or distilled oil.

Analytical identity is equally important. Four *A. herba-alba* oils showed variable enantiomeric proportions of α-thujone, β-thujone, and camphor when chiral methods were applied (Said Mel et al. 2016). In *A. persica*, structure-confirmed monoterpenoids and bisabolane sesquiterpenoids expanded known chemical space, but limited activity in the reported tests did not establish antiparasitic function (Umaraliev and collaborators 2025, 2026). The comparative dataset therefore retains presence/absence, relative abundance, and analytical detectability as separate variables.

Full-text cross-species assays quantify the same context dependence. Five Chinese *Artemisia* oils yielded 0.02–0.53% and separated by PCA/HCA, with species-specific dominant constituents and strongest contact phenotypes from *A. ordosica* or *A. tangutica* against two stored-product insects. Five Pakistani *A. scoparia* populations varied in oil yield and capillene/beta-myrcene/gamma-terpinene abundance; the high-altitude Swat oil remained *Aedes aegypti*-repellent for up to 165 minutes, while the Attock oil had larvicidal LC50 values of 55.5 and 43.5 microgram/mL at 24 and 48 hours. A camphor-rich *A. vulgaris* oil produced route- and time-dependent mite toxicity, but also contained a substantial non-terpene library assignment. These findings support species- and environment-stratified mixture experiments, not single-terpene causal claims (J. W. Zhang et al. 2022; Parveen et al. 2024; Kunnathattil et al. 2025).

5. Comparative evolutionary genomics: what is supported

5.1 A bounded comparative analysis

The Terpedia package includes a completed full two-proteome OrthoFinder run for *A. annua* and *A. argyi*, restricted pathway-family mapping, and a bounded three-species diagnostic adding *A. tridentata*. The two-proteome run assigned 111,235 of 129,761 proteins to 28,696 orthogroups, including 25,039 groups containing both species and 7,086 single-copy groups. These results demonstrate computational feasibility and identify candidate shared and lineage-restricted families.

They do not constitute a genus-wide evolutionary analysis. A two-tip species tree is necessarily trivial. The three-species pathway diagnostic is candidate-focused and depends on heterogeneous assemblies and annotations. Copy-number comparisons are sensitive to ploidy, homeolog retention, assembly collapse, gene prediction, and annotation release. Absence from an annotation is therefore not biological absence. The strongest current conclusion is that *Artemisia* terpene evolution can be investigated with an orthology-aware workflow, not that comparative genomics has already explained antiparasitic diversity.

The 2025 global phylogeny supplies a broader taxonomic backbone for this unfinished analysis. Its gigamatrix placed 394 of 505 accepted *Artemisia* species from 202 low-copy nuclear genes plus ITS/ETS, with genome-skimming data for 314 species and extensive nuclear/plastid discordance (Jiao et al. 2025). This enables phylogeny-stratified selection of *A. annua*, *A. absinthium*, *A. argyi*, *A. herba-alba*, *A. cina*, and other chemistry-bearing taxa. It does not establish terpene-gene orthology, duplication/loss, pathway flux, or antiparasitic adaptation; those questions still require common, ploidy-aware nuclear annotation and reconciled gene trees.

5.2 From orthogroups to testable gene–metabolite hypotheses

The useful evolutionary unit is a linked chain rather than a gene list:

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species tree → gene tree/orthology → tissue expression → enzyme product
      → native metabolite → preparation exposure → parasite phenotype
      → host selectivity and target engagement
```

Each arrow is a potential failure point. A candidate TPS must be tested with a defined substrate and authentic product standards. A P450 candidate needs substrate-matched biochemical assays and tissue context. A metabolite association requires quantitative measurement in the same material used for the parasite assay. A parasite target hypothesis needs orthogonal confirmation rather than docking alone.

Table 3 separates the comparative-genomics claims supported by the current package from the experiments still needed to discriminate them.

Table 3. Current comparative-genomics claims and the next discriminating experiment.

Claim level	Supported now	Not established	Discriminating experiment
Shared pathway ancestry	Candidate orthogroups and conserved pathway-associated sequences	Orthology across the genus and ancestral state	Common reannotation, species tree, gene-tree reconciliation
Gene-family expansion	Candidate TPS/P450/OSC differences in sampled genomes	Causal expansion or lineage-specific innovation	Chromosome-scale assemblies, synteny, homeolog-aware copy number
Catalytic divergence	Recombinant *Artemisia* enzyme and TPS evidence in selected cases	Native product and ecological function across species	Purified orthologs, isotope-labeled substrates, authentic standards
Chemotype evolution	Population, tissue, stage, and environmental associations	Heritable basis independent of environment	Common-garden multi-omics and QTL/allele-aware analysis
Antiparasitic evolution	Artemisinin pathway and defined assay anchors	Genus-wide gene-to-parasite causality	Matched chemotype, fractionation, target engagement, host counterscreens

The most productive next comparison is not "which species has more TPS genes?" It is whether pathway completion, tissue localization, and product spectra predict a defined antiparasitic phenotype after preparation is standardized. This is especially important for the distinction between *A. annua* artemisinin chemistry and *A. absinthium* thujone, STL, and historical santonin chemistry.

The linked evidence chain is shown in Fig. 1.

6. Antiparasitic evidence

6.1 Evidence ladder and protein interactions

The interaction map separates direct target evidence, functional target-class evidence, enzyme or activity-based evidence, computational hypotheses, and unresolved targets. Artemisinin has the strongest direct malaria anchor, with heme activation, redox chemistry, proteome-wide or chemoproteomic signals, and parasite-state resistance studies contributing complementary rather than identical evidence. Calcium-homeostasis evidence extends to *Toxoplasma gondii* and related systems; recombinant enzyme assays for non-*Artemisia* STLs provide biochemical plausibility; docking against *Leishmania*, *Haemonchus*, or insect proteins remains hypothesis-generating unless paired with biochemical or genetic validation.

This separation prevents a common category error: treating a docking score as equivalent to target engagement, or treating an oil phenotype as evidence that one named constituent binds the proposed target. The 19-record interaction table is consequently a map of evidence gaps as much as a list of targets. For most non-artemisinin terpenes, the correct statement is that a preparation or purified compound has a phenotype and that target identity remains open.

To make this critical comparison reproducible, we applied the provisional non-compensatory rubric in `evidence-appraisal-protocol.md` to all 129 antiparasitic records. Twenty-one records had defined or purified chemistry, 36 had a profiled mixture, and 72 retained crude or unresolved attribution; 86 reported direct in-vitro parasite/vector assays, 39 included an in-vivo or organismal endpoint, and four were indirect or surrogate records, including three hemozoin assays. Explicit host/selectivity controls were present in 22 records, partial controls in 24, and absent or unresolved in 83. Only three records reached the E4/E5 functional-target-class level through the linked interaction map; 23 were hypothesis/stage-mechanism level and 103 were phenotype-only or unresolved. The overall appraisal therefore yielded 24 B-level characterized phenotypes and 105 C-level phenotypes with unresolved attribution; no record met the deliberately strict A tier. This is a limitation of evidence resolution, not evidence that the preparations lack activity. Row-level judgments and their rule basis are archived in `evidence-appraisal.csv`.

6.2 Malaria: artemisinin as the validated anchor

Artemisinin is a sesquiterpene lactone with an endoperoxide bridge whose activation is central to its antimalarial pharmacology. The pathway and the parasite mechanism should be kept conceptually separate: a plant enzyme establishes production, while parasite experiments establish exposure, activation, damage, resistance, and selectivity. Artemisinin-related chemistry has been studied across asexual blood stages, gametocytes, resistant strains, rodent malaria, and whole-plant or tea preparations. These layers are not equivalent, but together they provide a much stronger evidence base than is available for most *Artemisia* terpenes.

The heme anchor is quantitatively reproducible in the full-text record. In synchronized 10–15-hour ring-stage *P. falciparum* TM267R, artemisinin IC50 fell from 7.0 +/- 1.1 nM to 3.1 +/- 0.3 nM with 0.1 micromolar hemin and 1.9 +/- 0.5 nM with 1 micromolar hemin. Severe-malaria plasma heme was 3.5 +/- 0.7 micromolar versus 0.3 +/- 0.2 micromolar in normal subjects; hemin increased TBARs in parasite systems, and vitamin E attenuated the oxidative signal. Quinoline comparators were not potentiated. This supports a purified-artemisinin heme/redox model in a synchronized blood-stage assay, not a sole parasite receptor, a crude-plant mechanism, or an infected-host treatment effect (Tangnitipong et al. 2012).

Whole-leaf and tea studies are especially informative about matrix effects and especially limited for constituent attribution. Mutant or chemically contrasted *A. annua* materials, tea infusions, and extract/isolated-artemisinin comparisons have reported activity that cannot always be predicted by artemisinin concentration alone. A gametocyte study compared *Artemisia* infusions with artemisinin, and a multistage study reported inhibition across parasite stages including relapse-relevant hypnozoite models (Balaich et al. 2016; Snider and Weathers 2021; Ashraf et al. 2022). These results support matrix-aware pharmacology, not a general claim that crude tea is equivalent to an approved artemisinin combination treatment.

Recent metabolomics sharpens the matrix question. Shi et al. used UPLC-Q-TOF-MS/MS to compare wild and cultivated *A. annua*, reporting 24 flavonoids, 68 sesquiterpenoids, and 21 other compounds and nominating candidate non-artemisinin and arteannuin-B-related chemistry in a discovery and fractionation framework (H. Shi et al. 2022). The study identifies plausible contributors and prioritizes fractionation; it does not establish that a nominated sesquiterpene is the causal antimalarial molecule.

A full-text hydrotrope study provides a controlled matrix example. Cryoground leaves were extracted with optimized aqueous sodium- or cholinium-salicylate solutions, yielding 6.50 +/- 0.10 and 6.44 +/- 0.09 mg/g artemisinin. At matched 1 microgram/mL artemisinin-equivalent exposure against susceptible *P. falciparum* 3D7, the best extracts had IC50 values of 0.0027 +/- 0.0005 and 0.0034 +/- 0.0009 microgram/mL versus 0.0066 +/- 0.0025 microgram/mL for pure artemisinin; arteannuin B/J and several flavonoids co-occurred. The extracts were reported as nonhaemolytic at the closest IC50 dose in cultured erythrocytes, but the single-line, mixed-chemistry design did not establish a causal terpene, in-vivo selectivity, or clinical safety (Ferreira et al. 2024).

A defined-component malaria study adds a complementary matrix and host-exposure anchor. Artemisinin, arteannuin B, arteannuinic acid, and scopoletin were combined at 1:1:1:1 by mass; in four-day *P. yoelii* and *P. berghei* mouse experiments, the authors

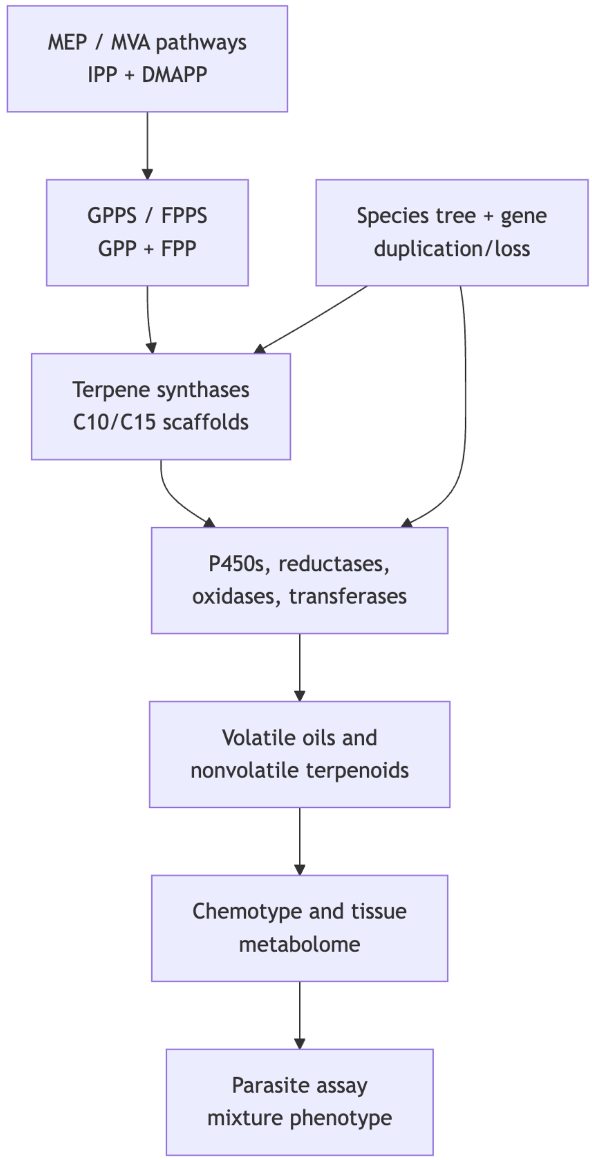

Fig. 1 Conceptual route from precursor supply to phenotype

reported approximately 93% parasitemia reduction for the combination versus approximately 31% for pure artemisinin. Repeated-dose LC-MS/MS pharmacokinetics also showed higher artemisinin AUC and Cmax in healthy and infected mice. The result is preclinical and preparation-specific: scopoletin is a non-terpene coumarin, single-component causality was not factorially resolved, no parasite protein was directly engaged, and the study does not establish whole-plant or clinical efficacy (J. Li et al. 2018).

A publisher-indexed full-text study used a top-down PD–PK design to compare ethyl-acetate and petroleum-ether *A. annua* extracts against pure artemisinin. In *P. yoelii* mice, reported ED50 values were 5.6 mg/kg for pure artemisinin, 2.9 mg/kg for the ethyl-acetate extract, and 1.6 mg/kg for the petroleum-ether extract; the extracts did not enhance activity in the reported *P. falciparum* in-vitro assay. Petroleum ether increased rat artemisinin AUC0–t 1.7-fold, and differential LC-HRMS analysis isolated the sesquiterpenes arteannuin B and artemisinic acid, present at higher proportions in petroleum ether than ethyl acetate. Same-proportion testing supports an absorption/matrix hypothesis, but the experiment did not provide a complete factorial causal test, direct parasite-protein engagement, or a clinical botanical formulation (Cai et al. 2024).

Aubouy et al. compared teas from ten Benin plantations with French material using LC-HRMS and *P. falciparum* 3D7 assays. Artemisinin ranged from 0.3 to 15.7 mg/L, storage produced substantial losses in several Benin locations, and activity correlated positively with artemisinin while comparison with pure artemisinin suggested enhancing or limiting co-metabolites (Aubouy et al. 2025). This is strong origin-, storage-, and matrix-level evidence. It is not terpene-specific mechanism, standardized product performance, or clinical efficacy.

A full-text postharvest chemistry control further shows why a malaria-associated metabolomic feature cannot automatically be treated as an active terpene. Voucher-linked Beninese leaves and stems yielded the previously undescribed cadinapyridine sesquiterpene alkaloids annuanines A and B, with annuanine A confirmed by X-ray diffraction. Both compounds were inactive against chloroquine-resistant *P. falciparum* FcB1 at the reported concentrations and lacked notable activity in the tested Caco-2, VERO, and Thp1 cell models. Annuanine B was absent from fresh-plant tea and appeared after drying/storage; mixture experiments attributed activity to artemisinin rather than annuanine B. The proposed biosynthetic versus extraction/degradation origin remains unresolved, but the study provides a defined-compound negative control and a direct reason to include storage and artifact controls in untargeted *Artemisia* metabolomics (Fabre et al. 2025).

Full-text verification of two *A. afra* malaria studies sharpens the species boundary. A Cameroon acetone-extract comparison measured 19.3 mg/g artemisinin in dry *A. annua* extract, *A. afra* below the reported HPLC-UV detection limit, and LDH IC50 values of 0.09 +/- 0.01 versus 11.73 +/- 2.15 microgram/mL. Late-stage *P. falciparum* metabolomics placed *A. annua* close to purified artemisinin, whereas *A. afra* shifted folate, carbohydrate, phospholipid, amino-acid, and lipid-precursor metabolism. A separate Senegal powder study found 93–98% day-3 activity for *A. annua* at 12.5–25 mg/kg artemisinin-equivalent doses but only 21–28% for single-dose *A. afra* and less than 5% after three daily high doses in *P. berghei* mice. Together these full-text controls support preparation- and host-dependent malaria interpretation; they do not identify a non-artemisinin terpene or falsify all tea-, origin-, or schedule-dependent *A. afra* hypotheses (Mamede et al. 2025; Walz et al. 2024).

The first full-text verification tranche makes these boundaries more concrete. Fresh-herb soaking or pounding changed artemisinin concentration from 14.5 mg/L in the dried-herb infusion to 293 mg/L in pounded juice, but only the pounded juice suppressed *P. berghei* parasitemia in the reported mouse comparison; the preparation effect therefore cannot be summarized as a universal tea benefit (Wright et al. 2010). In a separate origin comparison, Cameroon and Luxembourg infusions differed in chromatographic profiles and beta-haematin inhibition, with the reported 10% dilution absorbance of 0.075 versus 1.515, respectively; this is a heme-surrogate and chemotype/matrix result rather than direct parasite efficacy (Akkawi et al. 2014). A voucher-linked *A. annua* study normalized several extract assays by artemisinin and compared sensitive and K13-mutant *P. falciparum* lines; activity remained strain- and exposure-time-dependent, and pooled malaria-exposed sera produced additional inhibition, supporting a host/matrix hypothesis without identifying a causal non-artemisinin terpene (Gruessner and Weathers 2021). Finally, matched *A. annua* and artemisinin-free *A. afra* infusions inhibited blood and hepatic stages, including relapse-relevant models, but the active chemistry was not resolved to individual terpenes (Ashraf et al. 2022). These verified findings are archived row by row in `full-text-verification.csv`; they strengthen the evidence map while leaving constituent attribution and clinical translation open.

The newly verified 2013 comparison tested 16 geographically and temporally varied *A. annua* samples plus two *A. afra* samples as teas and chloroform extracts using HPLC-ELSD-normalized artemisinin measurements and a single chloroquine-sensitive *P. falciparum* 3D7 erythrocyte-stage assay. *A. annua* tea extracts averaged 0.75 microgram/mL extract IC50 and 4.97 ng/mL calculated artemisinin IC50 versus 5.48 ng/mL for pure artemisinin; chloroform extracts averaged 0.18 microgram/mL and 5.27 ng/mL, while *A. afra* teas were inactive above 25 microgram/mL and its chloroform extracts averaged 11.02 microgram/mL. The aged material, absent comprehensive terpene profiling, and single-stage design leave storage-sensitive constituents, prodrugs, and other life-cycle effects unresolved. The result supports artemisinin-dominant activity in the tested matrices, not the stronger genus-wide claim that artemisinin is the only active constituent (Mouton et al. 2013).

Two full-text chemistry controls further separate malaria evidence from terpene attribution. Authenticated *A. indica* stems were extracted with n-hexane and methanol, and activity-guided silica fractionation yielded exiguaflavanones A/B, maackiain, and a benzofuran. Against multidrug-resistant *P. falciparum* K1, exiguaflavanones A and B had EC50 values of 4.60 and 7.05 microgram/mL, respectively; these are defined non-terpene constituents, not evidence for a terpene or STL mechanism. In Cape Verdean *A. gorgonum*, leaves and flowers yielded guaianolides, a secoguaianolide, and germacranolides; ridentin showed FcB1 IC50 3.8 +/- 0.7 microgram/mL versus Vero-cell IC50 71.0 +/- 3.9 microgram/mL. The latter is a structure-resolved STL observation from one endemic species, but publisher-preview limits leave some per-compound values unresolved and do not establish an artemisinin pathway or cross-species transfer (Chanphen et al. 1998; Ortet et al. 2008).

The expanded verification set adds a more explicit plant-matrix pharmacology result. A petroleum-ether/silica extract quantified artemisinin, deoxyartemisinin, arteannuin I, dihydroartemisinic acid, and artemisinic acid as a sesquiterpene-rich mixture; at the lowest matched artemisinin dose, the extract produced approximately 42% murine inhibition versus approximately 4% for pure artemisinin, while rat exposure increased and Caco-2 assays implicated altered P-glycoprotein efflux. The study supports a preclinical matrix/host-transporter hypothesis, not a single-terpene parasite mechanism (Xu et al. 2025). A complementary genetic control found that removing major flavonoids while retaining wild-type artemisinin did not materially reduce *P. falciparum* Dd2 activity, whereas artemisinin-impaired lines lost activity; this strengthens artemisinin's role in that defined whole-leaf comparison without excluding matrix effects in other preparations (Czechowski et al. 2019). Silicon-induced trichome enlargement and higher infusion artemisinin likewise did not produce stronger *T. gondii* growth impairment, separating plant storage traits from parasite efficacy (Rostkowska et al. 2016).

Malaria resistance makes the evidence hierarchy more important. Artemisinin resistance is a parasite-state and genetic problem involving stress handling, proteostasis, heme detoxification, and redox homeostasis. A whole-plant or co-metabolite intervention could alter exposure or activity, but the responsible molecule and host selectivity must be established experimentally. The appropriate next study is a matched panel in which the same voucher-linked material is analyzed by targeted LC/GC methods, fractionated, tested against synchronized parasite stages and resistant lines, and paired with host-cell and heme/redox controls.

The newly verified malaria tranche includes both a resistance-evolution and a transmission-stage control. Orally delivered dried whole-plant *A. annua* reduced an artemisinin-resistant *P. yoelii* line more effectively than matched pure artemisinin and delayed stable resistance during serial *P. chabaudi* selection, but the study did not resolve a responsible terpene or establish a human regimen (M. A. Elfawal et al. 2015). In synchronized *P. falciparum* NF54 cultures, 5 g/L *A. annua* and *A. afra* teas reduced asexual parasites and gametocytes, with stronger effects tracking artemisinin content and only a weak artemisinin-deficient *A. afra* signal (Snider and Weathers 2021). These findings support a conditional artemisinin-centered, matrix-sensitive hypothesis, not generic terpene synergy or clinical transmission blocking.

Full-text verification adds a defined non-terpene resistance comparator. Infected male ICR mice received artemisinin plus the polymethoxylated flavonol chrysosplenetin at 40:80 mg/kg, with sensitive or resistant *P. berghei* K173 and Mdr1a wild-type or knockout backgrounds. Across the seven-day course, the combination retained 100% survival, approximately 30% inhibition, and approximately 20% parasitemia in resistant cohorts regardless of Mdr1a genotype, while tissue- and genotype-dependent ABC-transporter, ROS/GSH, cytokine, PI3K/AKT-mTOR, and MAPK responses changed. The study strengthens host/matrix and resistance-state hypotheses, but chrysosplenetin is not a terpene and no parasite-protein engagement or terpene-resolved botanical exposure was tested (Ji et al. 2025).

The following full-text checks add useful dose and mechanism boundaries. Powdered dried *A. annua* leaves containing 24 mg/kg artemisinin reduced *P. chabaudi* parasitemia more than equal pure artemisinin from 12 to 72 hours, but GC-MS-resolved leaves were not fractionated to identify a non-artemisinin terpene (M. Elfawal et al. 2012). Purified DHA, artemether, and artesunate inhibited beta-hematin crystallization only under reduced conditions in a controlled heme-detoxification assay, while a chemical-genetic *P. falciparum* study linked DHA survival to UPR resolution and proteasome capacity across Kelch13 and proteasome-mutant backgrounds (Ribbiso et al. 2021; Rosenthal et al. 2024). These are mechanistic anchors for artemisinin drugs, not proof of a whole-plant terpene mechanism. Full-text verification of an Iranian three-species screen adds a surrogate-only boundary. Authenticated aerial parts of *A. ciniformis*, *A. biennis*, and *A. turanica* were successively extracted across five solvent systems; the strongest result was *A. ciniformis* dichloromethane extract in synthetic beta-hematin formation (IC50 0.92 +/- 0.01 mg/mL). The authors' lactone interpretation was not supported by terpene-resolved chemistry, and no parasite culture, host-selectivity, or in-vivo endpoint was included (Mojarrab, Naderi, and Heshmati Afshar 2015).

The newly verified *A. armeniaca/A. aucheri* comparison provides a second, chemically unresolved malaria-surrogate anchor. Voucher-linked aerial parts were successively extracted across petroleum ether, dichloromethane, ethyl acetate, ethanol, and 1:1 ethanol-water, and dichloromethane extracts were silica-fractionated. In synthetic beta-hematin formation, dichloromethane extracts reached IC50/IC90 values of 1.36/2.12 mg/mL for *A. armeniaca* and 1.83/2.62 mg/mL for *A. aucheri*, while the strongest *A. armeniaca* fraction reached 0.47/0.71 mg/mL. TLC detected mixed terpenes, fatty acids, methylated coumarins, and methoxylated flavonoids, but no compound-resolved terpene, artemisinin, or sesquiterpene-lactone measurement was reported. Because the assay was synthetic beta-hematin rather than parasite culture, this result prioritizes fractionation without establishing blood-stage efficacy or a terpene mechanism (Mojarrab et al. 2014).

Full-text verification also adds a comparative wormwood-family nematode control. Cold-methanolic *A. herba-alba* flower and aerial-part extracts produced concentration- and time-dependent *H. contortus* adult-motility effects and 98.67% and 88.3% egg-hatch inhibition, respectively, at 1 mg/mL. Qualitative phytochemical screening—not terpene-resolved profiling—was used, and no mammalian selectivity endpoint was reported (Ahmed et al. 2020). This supports preparation-specific veterinary activity while leaving transfer to *A. absinthium*, thujone/STL attribution, in-vivo efficacy, and human safety unresolved. The newly verified goat coccidiosis study provides a negative translation control: a single 2 g/kg crude ethanolic *A. absinthium* dose was less effective than toltrazuril or amprolium, with oocysts persisting through day 28 (Iqbal et al. 2013). In a separate 46-plant tincture screen, 1.0% *A. absinthium* killed 85.5 +/- 16.5% of *Strongyloides papillosus* L1-2 larvae but only 19.4 +/- 6.7% of L3 larvae and none of *Haemonchus contortus* L3; no terpene profile or in-vivo endpoint was reported (Boyko and Brygadyrenko 2025).

Full-text verification of a Zimbabwean *A. herba-alba* study adds an adult-worm boundary: leaves were oven-dried at 40 degrees C, water-macerated for 48 hours, and tested at 1-10 mg/mL against 30 mature *H. contortus* worms per replicate. Approximately 80% mortality occurred at 5 mg/mL by 8 hours, and 10 mg/mL killed all worms by 6 hours, with albendazole as a positive control. No terpene-resolved chemistry, host-cell counterassay, infected-host endpoint, residue, or systemic exposure was reported, so this supports a wormwood-family veterinary phenotype rather than a thujone/STL mechanism or direct extension to *A. absinthium* (Hoshiki, Soul, and Bernard 2025).

A full-text *A. cina* formulation study adds a separate oxidative-stress class. Authenticated cultivated leaves and stems produced an aqueous extract used to synthesize quasi-spherical silver nanoparticles (7.56-nm TEM core; 57.9-nm DLS diameter). The particles produced 91.33% *H. contortus* L3 mortality at 125–500 microgram/mL, with reported LC50/LC90 values of 4.128/17.993 microgram/mL, while aqueous extract alone produced 21.54% mortality. SOD increased 2.67-fold at 8 hours and GPx/CAT decreased at 2 hours at a sublethal particle exposure. Because the capping metabolites were not terpene-resolved, AgNO3 was comparably active, and no mammalian or host efficacy endpoint was included, this is not evidence for native *Artemisia* terpene activity or a direct parasite-protein target (Hernández Guerrero et al. 2025).

Full-text malaria verification of a non-*annua* comparator further separates volatile composition from antimalarial causality. Flowering-stage Iranian *A. khorassanica* aerial parts were macerated in methanol, defatted, and partitioned; GC-MS of the non-polar fraction reported 29.9% monoterpenes, 6.2% sesquiterpenes, chrysanthenone at 7.8%, and cis-thujone at 5.8%. A crude extract dose was administered subcutaneously to *P. berghei*-infected NMRI mice for three weeks, with a limited microscopy-based parasitemia response. Because artemisinin was not quantified and no volatile was isolated or tested, this is a species- and preparation-specific murine comparator, not an *A. annua*-equivalent pathway or single-terpene mechanism (Nahrevanian et al. 2010). A related Iranian *A. sieberi* study used flowering aerial material from Khorassan and Semnan, yielding 13.3 g crude extract from 140 g plant material after 72 h methanol maceration, followed by n-hexane, chloroform, and ether partitioning. Total extract and the two semi-polar fractions reduced *P. berghei* parasitemia in NMRI mice, and the authors reported no major body-weight, survival, hepatomegaly, or splenomegaly signal at the tested exposures. However, the paper did not provide compound-resolved terpene chemistry, artemisinin measurement, exact effect sizes, stage-resolved IC50 values, or a parasite-protein target; it explicitly leaves the major active component for future work (Nahrevanian et al. 2012).

The gerbil study adds a full-text negative control to the wormwood section. Female gerbils received 600 exsheathed *H. contortus* L3 and were examined nine days later. Purified artemisinin (400 mg/kg once or 200 mg/kg for five days), *A. annua* aqueous or ethanolic extract (600 mg/kg), *A. annua* essential oil (300 mg/kg), and *A. absinthium* ethanolic extract (1,000 mg/kg) produced no significant reduction in worm burden. The *A. absinthium* group averaged 111.6 worms versus 97.1 in the water control; artemisinin groups averaged 72.6 and 78.2 versus 57.0 and 58.9 in vehicle and water controls. No terpene-resolved profile, systemic artemisinin/dihydroartemisinin measurement, Labrasol-only control, or parasite-protein assay was included. Because the model is short-lived and larval-stage restricted, the result should be compared with positive ovine wormwood studies by stratifying host, stage, preparation, bioavailability, and contact time; it does not show that all wormwood chemotypes lack vermifuge potential (Squires et al. 2011).

A separate full-text whole-plant *A. annua* sheep trial reinforces the distinction between plant content and delivered antiparasitic dose. Twenty-four *H. contortus*-infected Santa Inês sheep received preflowering, sun-dried top-third plant material at 0.2% or 0.4% body weight for 30 days; HPLC-IR quantified artemisinin at 0.96% of dry material. Mean efficacy was only 32.9% and 14.8% in the two plant groups, with non-significant log-EPG reductions and reduced intake at the bitter higher dose; levamisole averaged 89% efficacy. The study did not provide a complete terpene profile, systemic pharmacokinetics, residue data, mammalian selectivity panel, or evidence that artemisinin caused the effect. It is therefore a useful negative-to-modest comparator for malaria-centered artemisinin reasoning and a reminder that nominal plant concentration, palatability, exposure, and parasite model must be reported separately (Silva et al. 2017). Full-text verification also adds a chemically resolved species control. Hydrodistilled flowering *A. absinthium* oil was dominated by myrtenyl acetate (47.10%), beta-thujone (11.41%), and pulegone (8.50%), whereas *A. dracunculus* oil was dominated by methyleugenol, beta-asarone, and elemicin. At 2 microliter/mL, *A. dracunculus* oil caused 100.0% and 94.9% mortality of *Meloidogyne incognita* and *M. chitwoodi* J2s at 72 hours, while *A. absinthium* caused only 11.4% and 7.5% mortality and increased egg hatching relative to control. sPLS and docking prioritized candidate constituents but did not establish causal molecules or nematode targets. This preparation- and species-specific result argues against a generalized wormwood vermifuge inference and is not direct evidence for gastrointestinal helminths (Evlice et al. 2026).

The next verification tranche includes a purified sesquiterpene-lactone transmission benchmark. Parthenin exposure eliminated *P. falciparum* mosquito oocysts at 1,000 ng/mL after stage V treatment and reduced median oocysts by 83% at 100 ng/mL; parthenolide retained a 47% reduction at 10 ng/mL. Parthenin also reduced exflagellation and blocked *P. berghei* ookinete maturation. Because these are non-*Artemisia* compounds, with stated parthenin cytotoxicity concerns and no mammalian therapeutic-window endpoint in the paper, the result motivates stage-resolved screening of authenticated *Artemisia* STLs but cannot be relabeled as wormwood activity (Balaich et al. 2016).

The full-text goat study supplies a particularly informative translation boundary. Market-purchased Peruvian leaves were oven-dried and extracted with 70% ethanol; the bulk extract caused 82.2% larval mortality and 81.79% motility inhibition against *Trichostrongylus* L3 at 200 mg/mL after 8 hours, yet 600 mg/kg per day for three days produced no significant fecal-egg-count reduction in five naturally infected goats. The material was not chemically resolved for thujone, santonin, or sesquiterpene lactones. This positive-in-vitro/negative-in-vivo pattern supports bioavailability, rumen-metabolism, dose, or preparation hypotheses—not a uniform claim that wormwood is either efficacious or inactive (Cala et al. 2014). A publisher-full-text Awassi-sheep study adds a contemporaneous positive veterinary result: a 95% ethanolic *A. absinthium* extract, prepared by repeated 16-hour maceration and given once at 1 g/kg, produced 66.79% FECR on day 7 and 76.77% on day 15 in six sheep, versus 100% for albendazole. Because the material had no lot-specific terpene fingerprint and the trial had 18 animals with short follow-up, this is a preparation-specific ovine signal rather than evidence of a causal thujone/STL mechanism or human safety (Irehan et al. 2026).

Two publisher-full-text checks add a defined preparation rationale and a C15 constituent lead. In the *Hymenolepis nana* study, authenticated but store-purchased Egyptian aerial parts were shade-dried and extracted in boiled water for 72 hours; the aqueous route was chosen to reduce thujone exposure. Oral extract at 400 and 800 mg/kg for three days reduced mouse EPG by 61.8% and 98.8% and worm burden by 56.5% and 98.5%, respectively, while praziquantel produced 100% reductions in the reported comparison. Adult worms exposed to 1 or 5 mg/mL showed dose-dependent paralysis, death, tegumental injury, lipid accumulation, nephridial-canal damage, and intrauterine-egg degeneration. The result is a strong crude-extract anticestodal signal with an explicit safety-motivated extraction choice, not a thujone/STL efficacy estimate or human vermifuge recommendation (Beshay 2018). In authenticated pre-flowering *A. cina*, bio-guided fractionation of a 3.86%-yield ethyl-acetate extract isolated the low-yield sesquiterpene cinic acid (0.01% of extract). The extract had LC50/LC90 values of 2.56/3.30 mg/mL against *H. contortus* L3, while the paper reports stronger cinic-acid activity in its isolated-compound assays; assay-dependent comparisons and the low yield require independent standardization and reconstitution. This provides a C15 constituent-to-helminth template, but not evidence that cinic acid occurs in *A. absinthium* or engages a defined parasite protein (Arango-De la Pava et al. 2024).

The full-text *A. cina* study provides a defined-mixture model for the wormwood hypothesis. Authenticated pre-flowering leaves and stems were macerated in ethyl acetate and fractionated by silica chromatography; GC-MS, HPLC-DAD, and multidimensional NMR identified peruvin as a pseudoguaianolide sesquiterpene lactone and cinic acid as a related C15 sesquiterpene. In *H. contortus* L3 assays, peruvin plus 1-nonacosanol/hentriacontane produced 97.33% observed lethality versus 19.96% expected at matched 0.50LC25/0.50LC25 proportions (ratio 4.87), while peruvin plus cinic acid produced 40.42% versus 28.39% expected (ratio 1.42); the hydrocarbon/alcohol pair was additive for egg-hatching inhibition, and cinic acid alone showed no egg-hatching inhibition at the tested concentrations. The authors' protein-thiol and membrane-fluidity explanations remain hypotheses: there was no mammalian counterassay, infected-host test, exposure measurement, or direct parasite-protein experiment. This is therefore a strong comparative template for testing wormwood fractions, not evidence that peruvin or cinic acid occurs in *A. absinthium* (Arango-De-la Pava et al. 2024).

An adjacent full-text *A. cina* study provides a useful non-terpene comparator. Bio-guided n-hexane fractionation of authenticated pre-flowering leaves and stems, followed by NMR and positive-mode ESI-MS, assigned the active C2F5 subfraction as a 63:37 mixture of 3'-demethoxy-6-O-demethylisoguaiacin and norisoguaiacin. The crude n-hexane extract caused 80 +/- 9% *H. contortus* L3 mortality at 4 mg/mL; active C1F5 and C2F5 fractions reached 100% mortality at 2 and 1 mg/mL, respectively. Because the study tested a lignan mixture rather than purified individual standards, lacked terpene-resolved profiling, and reported no mammalian selectivity, infected-host, exposure, residue, or direct parasite-protein endpoint, it is retained as a fraction-to-phenotype comparator—not evidence for a terpene, thujone, santonin, or sesquiterpene-lactone mechanism (Higuera-Piedrahita et al. 2023).

The thirty-third full-text tranche provides full-text extraction of the Tunisian *A. campestris* study as a flavonoid-dominated comparative control. Aqueous and 95% ethanolic aerial-part extracts were tested against *H. contortus* eggs and adults: both reached complete egg-hatch inhibition at 2 mg/mL, with LC50 values of 1.00 and 0.827 mg/mL, respectively; at 2 mg/mL, ethanolic extract caused 91.30% adult-worm mortality at 8 hours and 100% at 24 hours, versus 3.22% and 70.96% for aqueous extract. HPLC-MS assigned quercetin and apigenin derivatives rather than volatile terpenes or STLs, and no host, PK, residue, or in-vivo endpoint was included. This is a solvent-, dose-, and stage-specific *A. campestris* phenotype, not a causal terpene/STL or *A. absinthium* result (Akkari et al. 2014).

The thirty-fourth full-text tranche adds cross-species fluke controls. An ethyl-acetate *A. ludoviciana* subsp. *mexicana* extract reached adult-*Fasciola hepatica* LC50/LC90/LC99 values of 80.1/310.9/362.9 mg/L and greater than 60% egg ovicidal action above 300 mg/L, but current-experiment chemistry was not terpene-resolved. An authenticated ethanolic *A. vulgaris* extract reduced *F. gigantica* hatching and caused concentration- and time-dependent adult-fluke tegumental damage, while UV-Vis chemistry measured broad tannin/flavonoid/saponin/alkaloid classes rather than a causal terpene or STL. Both are comparative in-vitro crude-extract phenotypes, not direct *A. absinthium* vermifuge or parasite-protein evidence (Ezeta-Miranda et al. 2023; Nurlaelasari et al. 2023).

6.3 Wormwood: vermifuge evidence without over-attribution

Full-text extraction of a Pakistani cross-species screen found that methanolic extracts of authenticated *A. indica* and *A. roxburghiana* inhibited mixed gastrointestinal nematode eggs, larvae, and adults in a concentration-dependent assay. At 50 mg/mL, egg-hatch inhibition was 85 +/- 21.2% and 80 +/- 28.3%, respectively; the study reported all tested adult worms dead at that concentration within 24 hours. The material was not terpene-profiled, and no purified active, infected-host efficacy, PK, residue, or mammalian selectivity endpoint was reported. This supports a species- and preparation-specific *Artemisia* nematode phenotype, not a thujone/STL mechanism or a direct *A. absinthium* inference (Khan et al. 2015).

The foundational ovine study compared crude aqueous and ethanolic aerial-part extracts of *A. absinthium* with albendazole. Both preparations impaired adult *H. contortus* motility in vitro, with the ethanolic extract faster and more active; day-15 fecal-egg-count reduction was 82.85% at 1.0 g/kg and 90.46% at 2.0 g/kg for the ethanolic extract, versus a maximum of 80.49% for the aqueous extract at 2.0 g/kg (Tariq et al. 2009). The study did not chemically resolve thujone, santonin, or sesquiterpene-lactone exposure, and its mixed gastrointestinal-nematode endpoint cannot be reduced to *H. contortus* alone. The result is important preparation- and polarity-specific ovine evidence, not proof of a named wormwood terpene, a standardized dose, or human safety.

An additional full-text wormwood study shows why chemotype and parasite identity must remain coupled. Four hydrodistilled oils from two cultivated Spanish *A. absinthium* populations were thujone-free and dominated by (Z)-epoxyocimene (25.89-42.72%) and chrysanthenol (12.71-30.32%). Ex-vivo oil treatment reduced *Trichinella spiralis* L1 infectivity by 72-100% at 0.5-1 mg/mL, and selected oils reduced intestinal larval counts by 49-66% at 100-500 mg/kg in mice; J774 macrophage growth inhibition remained below 20% at 200 microgram/mL. The same oils caused only 5.44-11.56% mortality of *Meloidogyne javanica* J2 at 1 mg/mL. Because no major component was established as causal, these findings support a thujone-free, chemotype- and parasite-specific oil hypothesis rather than attribution to epoxyocimene, chrysanthenol, or "wormwood" in general (García-Rodríguez et al. 2015).

Full-text extraction of a second *A. absinthium* rat study clarifies the wormwood life-cycle boundary. Voucher-linked flowering aerial parts from Antalya were shade-dried, Soxhlet-extracted with methanol at 64 degrees C for three days, redissolved in water, filtered, and lyophilized. Wistar rats received 300 mg/kg for 20 days beginning five days after *Trichinella spiralis* infection or 600 mg/kg for five days beginning day 45, with water-only infected controls and tissue larval counts by trichinoscopy and artificial digestion. The extract reduced larval rates by 37.7-63.5% across sampled muscles in the enteral phase and 43.7-59.9% in the parenteral phase; *A. vulgaris* was more potent overall. No acute toxicity was observed at the tested doses, but no positive anthelmintic comparator, PK/residue endpoint, or terpene-resolved chemistry was reported. The result is a crude, host- and stage-specific phenotype, not evidence for thujone, santonin, absinthin, or another causal terpene/STL (Caner et al. 2008).

Wormwood's vermifuge history has multiple chemical and taxonomic layers. Santonin is a historically important anthelmintic STL associated with *Artemisia* taxa used for that purpose, but historical santonin exposure should not be treated as evidence that modern *A. absinthium* tea, oil, or powder has the same pharmacology. *A. absinthium* contains volatile thujones, camphor, 1,8-cineole, borneol, sesquiterpenes, STLs, phenolics, sterols, and other matrix constituents. A preparation-level helminth phenotype therefore cannot be assigned to "wormwood terpenes" without chemical matching and fractionation.

The modern evidence is real but heterogeneous. Methanolic *A. absinthium* was among the more active preparations in a multi-plant *Haemonchus contortus* egg and larval comparison, while UPLC-MS/MS documented selected secondary metabolites; the result supports preparation-level activity rather than a terpene-specific mechanism (Váradyová et al. 2018). An *Ascaris suum* study found concentration-dependent effects of alcoholic *A. absinthium* extract on embryogenesis and larvae, but the authors called for deeper phytochemical attribution (Băieș et al. 2022). A recent *A. brevifolia* abstract adds a contemporary multi-stage comparator: crude ethanolic and aqueous extracts were tested against gastrointestinal-nematode eggs, larvae, and adult worms; LC-MS identified 33 compounds and reported sesquiterpene lactones among the chemical classes. It reported 87.5% egg-hatch inhibition for the ethanolic extract at 50 mg/mL and complete adult-worm motility inhibition after 8 hours at 25 mg/mL, versus 82.7% and 63.3% for the aqueous extract at the corresponding endpoints. Because full methods, constituent tables, larval dose-response, and host-selectivity data remain unavailable, this is abstract-level, preparation-specific evidence rather than proof that a named *A. brevifolia* terpene or lactone causes the phenotype (S. Hussain et al. 2025). A full-text naturally infected-cat study adds a small, chemically unresolved *Toxocara cati* signal: diethyl-ether flower extract given orally for seven days reduced fecal egg counts in some cats, but group sizes were three to five, no adult-worm burden or positive comparator was used, and a 0.3/0.6 mg/mL egg-development assay did not differ from control. Serum biochemical measures remained within the authors' stated physiological ranges, which supports short-term tolerability in that experiment but not clinical safety (Yıldız et al. 2011).

Two queued abstract-level comparisons add quantitative context to the wormwood vermifuge hypothesis. A beta-pinene-rich *A. campestris* essential oil contained 50 identified compounds representing 99.98% of the oil; it produced 100% *H. contortus* egg-hatch inhibition at 2 mg/mL (IC50 0.93 mg/mL), 66.6% adult-worm motility inhibition at 0.5 mg/mL, and 72% total-worm-count reduction in *H. polygyrus* at 5000 mg/kg. Because 2-undecanone was the second major constituent and is not a terpene, the result does not identify beta-pinene as causal. In the 13-plant comparison, methanolic *A. absinthium* stem extract was among the most effective against *H. contortus* eggs and larvae and contained 6.10 mg rutin/g dry matter; aqueous extracts showed a broader pattern of stronger ovicidal activity. Both records support preparation- and polarity-specific veterinary hypotheses, not terpene/STL attribution or human efficacy (Abidi et al. 2018; Váradyová et al. 2018).

A 2026 *A. maritima* field study adds a related-species whole-powder comparator: in naturally infected Kashmir sheep, 3 g/kg produced 70.40% day-21 fecal-egg-count reduction versus 85.81% for fenbendazole, with improved selected hematological measures. The available report does not provide terpene/STL composition, controlled single-parasite burden, residue limits, or transfer to *A. absinthium* (T. Wani et al. 2026).

An independent primary conference-abstract assay provides a defined-compound negative control. Artemisinin identified as extracted from *A. absinthium* did not change adult *H. contortus* time-to-death at the tested concentrations, whereas pumpkin-seed oil accelerated death relative to saline and allicin was not statistically different from fenbendazole. Because the source is restricted to adult worms in vitro and does not report a terpene-resolved plant profile, artemisinin purity, exact concentrations, host selectivity, pharmacokinetics, or infected-host efficacy, it should be read as a stage- and preparation-specific negative result rather than evidence that crude wormwood extracts or other chemotypes lack vermifuge potential (Wetmore et al. 2019).

In experimentally infected lambs, wormwood extract showed strong in-vitro egg-hatch effects, yet group-level fecal egg counts did not differ significantly in the reported feeding trial (Mravčáková et al. 2020). A recent small Awassi-sheep study reported a 76.77% fecal egg-count reduction on day 15 for crude ethanolic extract compared with 100% for albendazole, but the 18-animal regional study requires independent chemical, dose, residue, and safety confirmation (Irehan et al. 2026). A low-input swine-farm study likewise reported antiparasitic effects after *A. absinthium* powder, but co-occurring interventions and complex powder chemistry prevent terpene attribution (Băieş et al. 2024).

Full-text verification of the lamb study confirms the central negative translation result: 24 lambs were assigned to four six-animal groups, wormwood stem was supplied at 1 g dry matter per day for 75 days, and the matched aqueous extract had *H. contortus* egg-hatch ED50/ED99 values of 1.40/3.76 mg/mL, yet treated-group fecal egg counts and terminal worm counts did not differ significantly from controls. The paper's chemical analysis emphasized phenolic fractions and reported 0.35 g/kg dry-matter total flavonoids; it did not provide a terpene-normalized exposure. A separate full-text *A. suum* study used a 70% ethanolic *A. absinthium* extract and detected sesquiterpene lactones among many metabolites, but reported only mixed-extract developmental inhibition. These records support a preparation-specific vermifuge hypothesis and a strong bioavailability/host-metabolism experiment, not a thujone- or lactone-specific claim (Mravčáková et al. 2020; Băieș et al. 2022).

A full-text *A. annua* sheep study provides a matched artemisinin translation control. Water, sodium-bicarbonate, ethanol, and dichloromethane extracts contained 0.28%, 0.60%, 4.4%, and 9.8% artemisinin, respectively; the sodium-bicarbonate extract used in vivo contained no artemisinin and had high antioxidant capacity. Ethanol and dichloromethane extracts completely inhibited egg hatch in vitro, but single-dose sodium-bicarbonate extract and artemisinin produced only nonsignificant day-15 EPG reductions of 19.3% and 27.8% in naturally infected sheep. Fecal artemisinin peaked at 126.5 microgram/g at 24 hours and fell to 49.0 microgram/g at 36 hours. The study therefore separates in-vitro preparation activity, artemisinin content, exposure, and ovine efficacy, without supporting a terpene-specific anthelmintic mechanism (Cala et al. 2014).

A separate full-text *A. absinthium* schistosome study adds a useful constituent-control boundary. A dichloromethane leaf-rinse extract caused 100% mortality of adult *S. mansoni* at 200 microgram/mL, and fractionation yielded artemetin and hydroxypelenolide; both isolated compounds were inactive at 100 micromolar. The related-species comparator parthenolide from *Tanacetum parthenium* was active, but that result cannot be transferred to wormwood. The comparison demonstrates why "sesquiterpene lactone" is not itself an attribution: species, scaffold, preparation, concentration, and host counterassay must remain linked (Almeida et al. 2016).

Oil and volatile evidence broadens the hypothesis but does not resolve it. *A. absinthium* oil has shown trypanocidal or trichomonacidal activity in defined in-vitro systems, and thujone-rich *A. sieberi* oil has shown contact activity against poultry red mites; these are relevant antiparasitic phenotypes but are not equivalent to gastrointestinal nematode efficacy or human vermifuge use (Martínez-Díaz et al. 2015; Tabari, Youssefi, and Benelli 2017). In *A. absinthium*, extraction-method comparisons against *Tetranychus urticae* showed different oil potencies and a candidate sesquiterpene association, but no single active molecule was established (Chiasson et al. 2001).

Two full-text antileishmanial controls sharpen the extract-versus-compound boundary. Standardized Tenerife-grown *A. annua* extracts containing 0.13%, 0.75%, or 1.21% artemisinin produced *Leishmania infantum* promastigote IC50 values of 782.25, 231.97 +/- 20.75, and 58.53 +/- 2.20 microgram/mL; the 1.21% extract had an amastigote IC50 of 69.32 +/- 12.79 microgram/mL and retained greater than 90% J774A.1 macrophage viability to 400 microgram/mL. An aqueous *A. sieberi* extract was more active than purified artemisinin against *L. major* promastigotes in a separate assay, with reported IC50 values of 25 versus 50 microgram/mL and low effects on uninfected mouse macrophages. Neither study resolved a complete terpene/STL profile or parasite-protein target, so both support standardized fractionation and selectivity experiments rather than a shared non-*annua* terpene mechanism (Morua et al. 2025; Heydari et al. 2013). The related full-text *A. kermanensis* study provides a cross-class attribution control: structure-confirmed purification yielded a geraniol-derived monoterpene derivative plus eupatilin and a trimethoxylated flavone. The fraction and isolated compounds showed *L. major* activity with much higher reported macrophage IC50 values, but the strongest amastigote result was flavonoid-associated. This supports chemical fractionation and host-selectivity testing without establishing a shared *Artemisia* terpene mechanism or transfer to wormwood (Soleimanifard, Saeedi, and Yazdiniapour 2023).

The best wormwood experiment is therefore not another crude-extract screen. It is a chemically fingerprinted, fractionated, dose-resolved assay against defined nematode life stages, with thujone and STL quantification, matched vehicle controls, mammalian cell or organoid counterscreens, and in-vivo pharmacokinetic and residue measurements. Only after that sequence could an evolutionary comparison test whether a *A. absinthium* chemotype or STL pathway predicts a reproducible vermifuge phenotype.

6.4 Other parasite systems as comparative controls

Full-text verification of a commercial *A. annua* oil against engorged *Rhipicephalus annulatus* females found dose-dependent mortality from 33.2% to 100% by 36 hours and complete egg-laying failure at the highest tested dilution, but the oil was not chemically profiled and cattle safety, residues, field performance, and constituent causality were not tested. This is a vector/ectoparasite comparator rather than evidence for malaria treatment or a single terpene (Pirali-Kheirabadi and Teixeira da Silva 2011).

Other *Artemisia* evidence includes *Leishmania*, *Trypanosoma*, *Schistosoma*, *Eimeria*, *Toxoplasma*, ticks, mites, mosquitoes, and gastrointestinal helminths. These records are useful as comparative controls because they expose stage, route, host, and formulation effects. Purified STLs can show defined protozoan phenotypes; oils can show vector or ectoparasite effects; extracts can show rodent or ruminant activity. None should be pooled into one "antiparasitic potency" estimate without common units, matched exposure, parasite-stage annotation, and host controls.

Full-text extraction illustrates why oil composition and formulation must be retained together. In a vouchered Tunisian comparison, *A. campestris* oil was beta-pinene-rich and had a promastigote IC50 of 44 microgram/mL, whereas *A. herba-alba* oil was dominated by camphor, 1,8-cineole, chrysanthenone, and thujones and had an IC50 of 68 microgram/mL; macrophage CC50 values were 124.4 and 160 microgram/mL, respectively, in the study's matched counterassay (Aloui et al. 2016). These are oil-level, promastigote-stage results rather than proof that beta-pinene, camphor, or thujone is causal. A formulated *A. absinthium* oil also produced intracellular-*L. amazonensis* activity, but the nanocochleate route and exposure cannot be generalized to wormwood vermifuge use (Tamargo et al. 2017).

The vector-control tranche is a separate translation category. Nanoliposomes containing oils from *A. annua*, *A. dracunculus*, and *A. sieberi* had species-dependent LC50 values against *Ae. aegypti* and *An. stephensi*, while *A. dracunculus* oil in a nanogel was more potent than its nanoemulsion against *An. stephensi* larvae; the studies characterized mixtures and formulations, not a malaria-parasite target or mammalian safety margin (Sanei-Dehkordi et al. 2023; Osanloo et al. 2022). Finally, antimalarial activity of *A. annua*-associated fungal endophyte extracts was traced to validated quinones such as emodin and physcion rather than plant terpenes, providing a useful negative attribution control (Alhadrami et al. 2021).

Negative or boundary records are particularly valuable. Some *Artemisia* preparations fail against a tested parasite, or show activity only in a formulation or one life stage. These observations help distinguish broad cytotoxicity from parasite selectivity and prevent publication bias from turning every reported plant effect into a positive mechanism.

7. Safety and translation boundaries

Preparation identity is a safety variable. Artemisinin, an *A. annua* tea, a distilled essential oil, a hydroalcoholic extract, and *A. absinthium* powder are not interchangeable exposures. Thujone requires explicit quantification because neurotoxicity risk varies with chemotype, dose, route, and product. Regional *A. annua* and *A. afra* oils differ in camphor-, cineole-, and thujone-associated composition, while volatile pharmacokinetic studies show tissue distribution followed by elimination for defined oils rather than a universal safety profile (Radulović et al. 2013; Oyedeji, Afolayan, and Hutchings 2009; Hou et al. 2021).

Host pharmacology is another translation gate. Artemisinin can affect host calcium-channel systems in experimental models, *Artemisia* teas can inhibit or induce drug-metabolizing enzymes depending on preparation, and reproductive studies in rodents raise preparation- and dose-specific concerns (Urban and Schaefer 2020; Kane et al. 2022; Abolaji et al. 2013). These findings do not define a human dose, but they show why parasite activity must be paired with host counterscreens, CYP/transporter testing, reproductive toxicology, and residue analysis.

The translation rule is simple: in-vitro activity is not a therapeutic dose; a mixture phenotype is not a single-constituent mechanism; a computational interaction is not target engagement; and increased plant production is not clinical efficacy. The evidence map retains these distinctions so that uncertainty is preserved rather than hidden in a pooled score.

The target-evidence hierarchy used in this review is shown in Fig. 2.

The living-review study-selection flow is summarized in Fig. 3.

8. Testable hypotheses and research agenda

The integrated evidence supports six experimentally discriminating hypotheses.

1. Pathway completion will predict antiparasitic chemistry better than total TPS count. Across a phylogenetically sampled panel, complete precursor-to-oxidized-product modules should explain defined metabolite abundance better than raw TPS

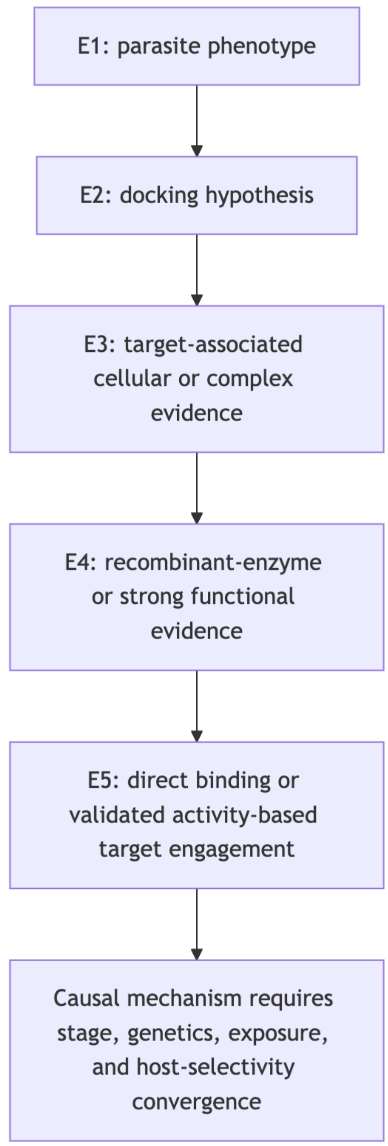

Fig. 2 Evidence ladder used for terpene–parasite target claims

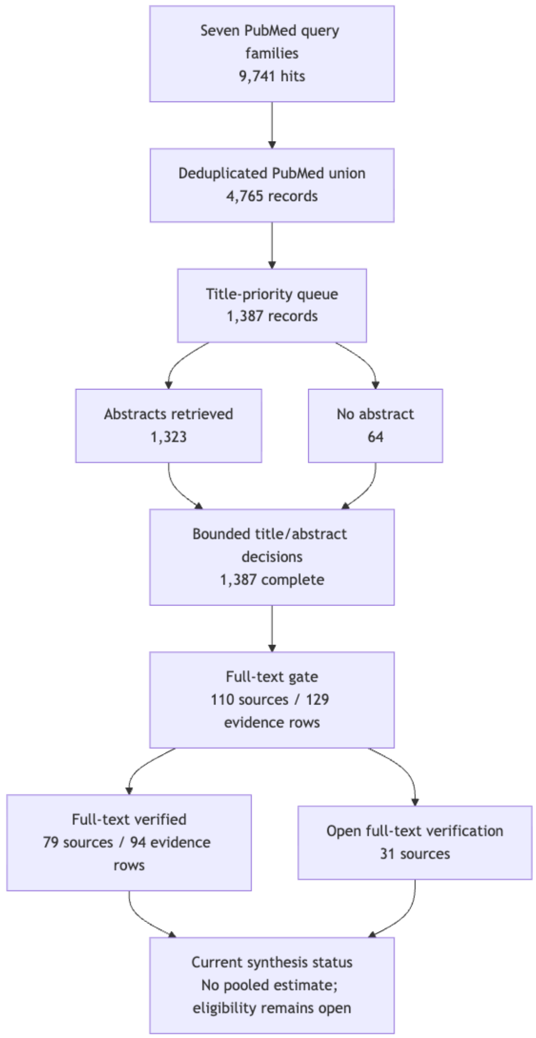

Fig. 3 Living scoping-review retrieval, screening, and full-text status

copy number. Test with common reannotation, gene-tree reconciliation, trichome expression, targeted metabolomics, and enzyme assays.

2. Secretory-tissue expression will outperform bulk-leaf expression. Cell-resolved or trichome-enriched transcriptomics should predict accumulated volatiles and STLs more accurately than bulk RNA-seq. Test with matched tissue sampling, spatial metabolomics, and isotope tracing.
3. Essential-oil antiparasitic activity will often be a chemotype-specific mixture property. Activity should be reproducible only when the full profile, stereochemistry, and exposure are matched. Test by reconstituting major and minor constituents at measured ratios and comparing whole oil, depleted oil, fractions, and purified compounds.
4. Artemisinin-like malaria phenotypes will depend on matrix and parasite state. Co-metabolites may alter uptake, metabolism, redox stress, or parasite-stage sensitivity without acting as standalone antimalarials. Test origin- and storage-matched teas against synchronized sensitive and resistant parasites, with LC-HRMS feature tracking and fractionation.
5. Wormwood vermifuge activity will require preparation-specific STL/volatile signatures. A reproducible nematode phenotype should track a chemical fingerprint and exposure window rather than the plant name alone. Test *A. absinthium* chemotypes against eggs, larvae, and adults, with thujone/STL quantification, host-cell counterscreens, and in-vivo residue studies.
6. Apparent gene-family expansion will be partly explained by cytotype and annotation. Homeolog retention, assembly collapse, and annotation choice should account for a measurable fraction of apparent TPS/P450/OSC expansion. Test chromosome-scale assemblies, allele-aware annotation, synteny, and reconciliation across a broad *Artemisia* species tree.

8.1 Minimum reporting standard

Future studies should report accepted taxon and voucher, accession or genotype, ploidy where known, tissue, developmental stage, locality, cultivation conditions, harvest and storage, preparation and extraction yield, analytical platform, authentic-standard or library status, stereochemistry, abundance units, parasite species and stage, inoculum, dose, endpoint, vehicle, positive control, host counter-screen, and raw-data availability. These fields are already represented across the Terpedia CSV schemas and should become the minimum unit for comparative synthesis.

9. Conclusions

There is sufficient primary research to support a strong *Artemisia* terpene review, but not to justify a single genus-wide potency score or a definitive evolutionary explanation of antiparasitic activity. The most secure conclusions are that *Artemisia* terpene chemistry is structurally broad and context-dependent; artemisinin biosynthesis and malaria pharmacology provide the best-resolved reference system; wormwood has a credible but preparation-specific vermifuge evidence stream; and comparative genomics currently provides testable prioritization rather than final causal explanation.

The central practical outcome is an evidence architecture. Chemical identity, tissue localization, pathway function, evolutionary history, parasite phenotype, target engagement, host selectivity, and translation must be linked without collapsing their uncertainty. Terpedia stores the full source registry, evidence matrix, screening records, interaction map, compound/specimen extraction, figures, code, checksums, and review-status record needed to extend this concise submission manuscript into a completed review.

Data and code availability

The complete auditable package is maintained in the Terpedia KB under the review-package directory and linked to the public `artemisia` source repository. The extended evidence-rich draft is retained as `article.md`; this file is the consolidated submission version. Large comparative-genomics inputs and outputs, search snapshots, structured evidence tables, claim audits, and SHA-256 manifests are versioned in the same package.

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Ethics approval and consent to participate

Not applicable. This review uses published literature and versioned public evidence records and reports no new human or animal experimentation.

Consent for publication

Not applicable.

Author contributions

Author names and CRediT roles must be supplied and approved by the submitting authors. The package separates literature retrieval, evidence extraction, comparative-genomics analysis, manuscript drafting, and critical revision as assignable roles.

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Use of artificial intelligence

An AI-assisted workflow was used during evidence organization, source-linked drafting, citation conversion, and editorial revision. The Terpedia registry, row-level evidence records, full-text verification fields, and rendered manuscript remain the auditable work products; human authors must independently verify the final text, approve authorship, and accept responsibility for the submission.

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